

TypeScript Frontend & Gateway – UML Class Diagram

UI5 Web Component Hierarchy • Observable Reactive Store • BackendGateway Driver Adapter Pattern

PACKAGE: core (Reactive Micro-Framework)

**<<abstract>>
Component**

```
# root: HTMLElement
- cleanupFns: Array<() => void>

+ mount(container: HTMLElement): void
+ unmount(): void
# requestRender(): void
# on(selector, event, handler): void
# track(cleanup: () => void): void
# template(): RawHtml
```

Store<T> (Observable)

```
- _state: T
- listeners: Set<(state: T) => void>

+ state: T
+ set(updater: (state: T) => void): void
+ subscribe(listener: () => void)
```

PACKAGE: views & dialogs (SAP Fiori Horizon UI5)

ListReportView

```
+ renderFilterBar(): RawHtml
+ renderTable(): RawHtml
- onRowSelect(guid: String): void
- onSort(column: String): void
- onSearchChange(term): void
```

ObjectPageView

```
+ renderHeader(): RawHtml
+ renderSummaryCards(): RawHtml
+ renderChatDrawer(): RawHtml
+ renderHistoryTab(): RawHtml
- onRollbackVersion(v): Promise
```

**<<abstract>>
DialogView**

```
# isOpen(): boolean
# requestClose(): void
# header(): RawHtml
# body(): RawHtml
# footer(): RawHtml
```

UploadDialogView

```
- selectedFiles: File[]
- progress: number

+ onDropFiles(e)
+ onStartIngestion()
```

BpmnModalView

```
- activeTab: BpmnTab
- xmlContent: String

+ switchTab(tab)
+ copyXml()
```

BackendSettingsModal

```
- currentMode: 'rest'|'mock'

+ onSwitchDriver(mode)
+ testConnection()
```

AuthModalView

```
- userProfile: UserProfile

+ onLogin()
+ onLogout()
```

PACKAGE: services/backend (Gateway & Adapter Driver Pattern)

BackendGateway (Facade)

```
- activeAdapter: BackendAdapter

+ dispatchBackend<T>(op, payload): Promise<T>
+ requestBackend<T>(op, payload): Promise<T>
+ setAdapter(adapter: BackendAdapter): void
+ getAdapter(): BackendAdapter
```

**<<interface>>
BackendAdapter**

```
+ testHealth(): Promise<BackendHealthResult>
+ getDocuments(params: DocumentQueryParams): Promise<DocumentListResult>
+ getDocument(guid: String): Promise<DocumentRecord>
+ uploadDocument(req: DocumentUploadPayload): Promise<DocumentRecord>
+ overwriteVersion(guid, req): Promise<DocumentRecord>
+ deleteDocument(guid): Promise<{success, message}>
+ regenerateSummary(guid, modelKey): Promise<SummaryRecord>
+ chatWithDocument(guid, req): Promise<ChatResponseResult>
```

RestBackendAdapter

```
- baseUrl: String ('/api/v1')
- buildHeaders(): Headers

+ fetchJson<T>(url, init): Promise<T>
+ uploadMultipart(url, formData): Promise<DocumentRecord>
+ implements all BackendAdapter methods via HTTP
```

MockBackendAdapter

```
- localDocuments: DocumentRecord[]
- simulateLatencyMs: number

+ filterInPlace(params): DocumentRecord[]
+ mockAiSummarize(guid, model): SummaryRecord
+ mockRAGChat(guid, prompt): ChatResponseResult
```